DN

DARIUS MIRCEA NISTOR

nistordarius26h@gmail.com | +40787726697 | **WWW:** www.linkedin.com/in/darius-nistor-3292783b1

Summary

Detail-oriented and analytical professional with expertise in Artificial Intelligence, data analytics, Python, cybersecurity, and business technologies. Experienced in transforming complex information into actionable insights, streamlining workflows, and applying modern AI tools to solve real-world challenges. Committed to continuous learning and professional growth, backed by certifications from leading global technology organizations and a strong passion for innovation, automation, and emerging technologies.

Skills

- Artificial Intelligence (AI)
- Data Analysis
- Python
- Generative AI
- Large Language Models (LLMs)
- Data Visualization
- Business Intelligence
- Cybersecurity
- Risk Management
- Amazon Web Services (AWS)
- Git & GitHub
- Open Source Software
- Process Optimization
- Critical Thinking
- Problem Solving
- Analytical Thinking
- Professional Communication
- AI Productivity Tools (Microsoft 365 Copilot)
- Web Technologies (HTML)
- Digital Marketing

Education and Training

Colegiul National De Informatica C.N.I. Moisil | Brasov
Some College (No Degree) in Mathematics And Computer Science

Languages

English:

Upper Intermediate (B2 - C1 EF SET Certified)

German:

Intermediate (B1)

Romanian:
Native

Certifications & Licenses

- Google AI Professional Certificate
- Google AI Essentials Specialization Certificate
- CS50's Introduction to Programming with Python (Harvard University)
- Career Essentials in Generative AI (Microsoft & LinkedIn)
- AI for Business Professionals (HP LIFE)
- Transformers in Practice - AMD (DeepLearning.AI)
- What Is Generative AI (LinkedIn)
- Open Source Foundations (IBM)
- Basics of Cyber Law (Cambridge International Qualifications / UniAthena)
- AWS Concepts Certification (Amazon Web Services)

•Claude Code 101 (DataCamp)

•Risk Management Course (The Open University & Rolls-Royce)

•Become an AI-Powered Marketer (Semrush)

•EF SET English Certification - C1 Advanced

•Introduction to HTML (CodeSignal)

Websites & Social Links

• www.linkedin.com/in/darius-nistor-3292783b1
• https://github.com/nistordarius26h-ship-it
• https://g.dev/NistorDarius

Projects

• Globally Accessible Autonomous Robot with AI-Based Tracking and Environmental Sensing
Designed and developed a fully remote-operable autonomous robotic platform, accessible from any location worldwide through cellular network connectivity. Engineered a distributed embedded architecture consisting of a low-level microcontroller unit handling real-time motor actuation, motion control, and sensor data acquisition, integrated with an onboard edge-computing module responsible for system orchestration, data routing, and remote communication. Developed a dedicated computer vision pipeline, executed on a separate high-performance workstation, to perform real-time object detection, tracking, and autonomous target-following behavior. Incorporated a diverse array of onboard sensors — including ultrasonic proximity sensors for collision avoidance, acoustic input for gesture/sound-based command triggering, battery voltage monitoring, environmental temperature and humidity sensing, and precipitation detection — to enable robust situational awareness. Integrated photovoltaic solar panels to provide sustainable, self-sufficient power for extended field deployment.

• Real-Time AI Object Detection and Tracking System
Designed and developed a computer vision system in Python for real-time object detection and multi-object tracking, capable of processing any live video stream. Integrated a deep learning detection model with a dedicated tracking algorithm to identify and persistently follow multiple object classes, including people, vehicles, and animals, while computing motion trails, real-time speed estimation, and occlusion handling to maintain tracking stability. Tested and validated using live footage from an aerial drone, the system was later adapted and integrated into an autonomous robotic platform to enable real-time person detection and autonomous target-following.

• AI-Based Aircraft Stall Margin Prediction System
Designed and developed a research prototype investigating the application of artificial intelligence to predict aircraft stall margins in advance of conventional stall warning systems. Explored the integration of forward-looking LiDAR sensing, onboard aircraft telemetry, and machine learning models to analyze airflow disturbance patterns and estimate stall risk proactively, supporting enhanced flight safety and data-driven decision-making.

• Automated Quantitative Trading System

Designed and developed a fully automated algorithmic trading system for the XAU/USD (Gold) market, engineered to operate without manual intervention. Built a quantitative trading strategy incorporating real-time market data analysis, signal generation, automated trade execution, and integrated risk management to support consistent, data-driven decision-making under live market conditions.

• ESP32 Wireless Security Research Platform

Designed a custom printed circuit board and developed a dedicated embedded research platform in Python to investigate Wi-Fi and Bluetooth communication protocols, wireless networking behavior, and embedded system security. Integrated custom hardware with purpose-built firmware to support controlled experimentation, protocol analysis, and vulnerability testing of wireless communication technologies.

• Portable Local AI Workspace

Configured and deployed a fully portable, offline-capable AI development environment using the open-source Odysseus workspace, hosted on a self-contained USB drive. Integrated and optimized local large language models (Gemma 3/4 and Qwen 3.5) to deliver a privacy-focused AI assistant capable of full inference without reliance on cloud infrastructure.

Awards & Honors

•InfoTron 2022 – 1st Place (Mechanical Engineering, Electronics & Computer Science Competition sponsored by Miele Brașov and Siemens Industry Software)
•InfoTron 2023 – 3rd Place (Mechanical Engineering, Electronics & Computer Science Competition)
•InfoTron 2026 – Participant

Activities

•Competitive Swimming Athlete (2012–2020) - Won 50+ regional and national medals, demonstrating discipline, resilience, teamwork, and performance under pressure.